標題 – 聚溫效應

由數據格點驅動的呼吸界面與粒子地景

Thermal Clustering Effect

摘要-Summary

本設計旨在將校園空間中無形的「群聚熱能」轉譯為可感知的物理地景。選定淡江大學建築系館旁之帶狀空間為實驗場域，透過感測器捕捉人群動態密度，並將數據導入演算系統。空間介入由「數據網格底座」與「應激反應單面」組成，將原本冷清的通行路徑定義為「能量緩衝容器」。當個體、雙人或群體進入場域，系統會根據能量熵值的變動，驅動鱗片單元進行 0° 至 90° 的動態翻轉。這不僅是物理空間的介入，更是一場關於數位邏輯與社會連結的空間實驗，讓隱形的能量流動最終凝結為場域中的新秩序。

This project aims to translate the invisible "collective thermal energy" within campus spaces into a perceptible physical landscape. The experimental site, located in the linear corridor adjacent to the Architecture Department at Tamkang University, utilizes sensors to capture real-time crowd dynamics and density, feeding this data into a generative system. The spatial intervention consists of a "data-grid base" and an "adaptive response interface," redefining a once-vacant transit path as an "energy buffer container." As individuals or groups enter the site, the system drives kinetic scale units to rotate from to based on fluctuations in entropy. More than a mere physical installation, this is a spatial experiment on digital logic and social bonding, allowing invisible energy flows to ultimately coalesce into a new field order.

Keyword

+熱能轉譯

+群聚效應

+能量波動

+應激界面

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設計操作說明

本設計的操作分為三個核心階段：數據編碼、參數對應與物理實踐。

首先是環境格式化：將基地簡化為黑白的點陣化二進位系統，建立絕對零度的背景。透過網格坐標定義出數據擷取的熱點，將現實中的走廊轉化為一個等待代碼啟動的數位底層。

其次是轉譯邏輯建立：利用 p5.js 演算邏輯模擬粒子流。當感測層捕捉到人體熱能（能量輸入），系統會計算其影響範圍。單人進入時產生局部擾動，多人群聚時則引發能量爆發，將抽象的「密度數值」映射為物理參數，如單元翻轉的角度 、反應速度與動作相位。

最後是應激構造執行：牆面與欄杆上的互動單元由微型馬達驅動。單元在 0° 閉合狀態下維持原本的場域秩序；當能量蓄積，鱗片開始旋轉，產生滲透的光影。在高熵值的群聚狀態下，單元達到 90° 全垂直開口，釋放隱藏的光源，形成如「集體星雲」般的發光地景。透過這種「呼吸界面」的重複與漸變，設計成功地將轉瞬即逝的社會行為，轉化為具備時空記憶的空間構造。

The operational logic of this design is divided into three core phases: Data Encoding, Parametric Mapping, and Tectonic Execution.

The first phase involves Field Formalization: the site is simplified into a high-contrast, black-and-white dithered system, establishing a state of "absolute zero." Through a coordinate grid, data acquisition hotspots are defined, transforming the physical corridor into a digital substrate awaiting algorithmic activation.

The second phase is the establishment of Translation Logic: utilizing p5.js algorithms to simulate particle flow fields. When the sensory layer detects human thermal energy (input), the system calculates the impact zone—categorized as Ambient, Diffusion, or Burst. A single person creates a localized disturbance, while a crowd triggers an energy surge. These abstract density values are then mapped onto physical parameters, such as the unit rotation angle , response speed, and action phase.

The final phase is Tectonic Execution: interactive units on the walls and railings are driven by micro-motors. In their closed state, the units maintain the original order of the field. As energy accumulates, the scales rotate to create permeable light and shadow. In high-entropy clustering states, the units reach a vertical opening, releasing internal light sources to form a glowing "Collective Nebula." Through the repetition and gradation of this "breathing interface," the design successfully transforms fleeting social behaviors into a spatial construct with temporal memory.